# Internship Report

**Course Code: SWE 4790**


**Submitted by:**

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**Performed at**

## TiCON Systems Ltd

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# Letter of Transmittal

February 17, 2026
Department of Computer Science and Engineering
Islamic University of Technology
Board Bazar, Gazipur-1704

**Subject: Submission of Internship Report for Evaluation**

Dear Sir,

I am pleased to submit my internship report as a part of the requirements for the Software Engineering program at the Islamic University of Technology (IUT). During my internship at TiCON System Limited, from October 12, 2025, to February 12, 2026, I had the opportunity to work as a Frontend Developer Software Engineering Intern. Under the mentorship of Mr. Shohanur Rahman, Senior AI/ML Engineer at TiCON System Limited, I gained hands-on experience in frontend development, particularly focused on the design, implementation, and optimization of software solutions for real-world applications.

Throughout my internship, I contributed to the development of frontend applications, gaining significant exposure to technologies like Next.js, TypeScript, and UI/UX design principles. This report outlines the objectives of my internship, the tasks I performed, the challenges I faced, and the technical and soft skills I acquired during this period. It also reflects on my contributions to the company's ongoing projects and how the experience has enhanced my academic and professional growth.

I respectfully submit this report for your review and evaluation, and I hope it fulfills the objectives and requirements of my internship as outlined by the department.

Thank you for your time and consideration.

Sincerely,
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# Acknowledgment

The author would like to express sincere gratitude to the entire team at TiCON System Limited for providing the opportunity to undertake an incredibly enriching internship experience. My deepest appreciation goes to Mr. MN Islam, the Chief Executive Officer and Founder of TiCON System Ltd., for his leadership and vision in creating such a dynamic and supportive work environment. His encouragement and open-door policy greatly contributed to making my time at the company both valuable and enjoyable.

I am particularly thankful to Mr. Shohanur Rahman, Senior Engineer from the AI/ML Team, who served as my internship supervisor and project manager. His expert guidance, patience, and willingness to share knowledge were fundamental in shaping my learning experience. His constructive feedback and mentorship were instrumental in refining my technical skills and improving my overall performance.

I would also like to extend my appreciation to all my colleagues at TiCON System Ltd. for their warmth, collaboration, and willingness to assist me throughout the internship. The team's camaraderie, openness to sharing ideas, and encouragement made my journey smoother and more rewarding. I am truly grateful for their support and guidance.

Additionally, I would like to acknowledge the Islamic University of Technology (IUT) for facilitating this wonderful opportunity. I am grateful to Junior Lecturer Aashnan Rahman, from the Department of Computer Science and Engineering, IUT, whose initiative and coordination played a vital role in organizing and facilitating my internship with TiCON System Ltd. His continuous support and advice ensured that I was well-prepared for the challenges I encountered during the internship.

Finally, I would like to extend my heartfelt thanks to all the professors and mentors at IUT, whose teachings and guidance throughout my academic journey have been indispensable in shaping my knowledge and skills, preparing me for this internship and future professional endeavors.

# Executive Summary

This document provides an overview of my internship journey at TiCON System Limited from October 12, 2025, to February 12, 2026, where I focused on refining my technical skills and gaining practical experience as a Frontend Developer. The internship gave me the opportunity to work with a range of industry-standard tools and technologies, specifically in frontend development, while contributing to projects aimed at creating user-friendly, scalable applications.

During my time at TiCON, I contributed to the development of the Learning Management System (LMS) frontend, working primarily with Next.js and TypeScript to build interactive and responsive user interfaces. I also had hands-on experience in implementing frontend components that are modular, reusable, and aligned with best practices in UI/UX design. These tasks significantly enhanced my proficiency with modern web development frameworks.

Furthermore, I worked on the full frontend development of the mock test application and also developed the frontend of the LMS, which involved creating dynamic features to improve user engagement. I implemented various user interface components and ensured that the design was intuitive and responsive. In addition, I was responsible for integrating the frontend with the backend API developed by my teammate, ensuring seamless data flow and a consistent user experience across platforms.

The internship experience also provided me with insights into the Agile development process, where I participated in sprint planning, daily standups, and code reviews. This exposure to industry workflows significantly improved my understanding of the software development lifecycle, particularly in the context of team collaboration and iterative development.

In addition to the technical skills I gained, the internship helped me develop valuable soft skills, including effective communication, teamwork, and time management. The experience taught me the importance of maintaining a high level of professionalism and discipline, especially in a hybrid work environment where remote collaboration played a significant role. I also learned the value of transparency and constructive feedback in team settings, which helped me continuously improve my work.

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# 1 Introduction

## 1.1 Background

An internship serves as a critical bridge between academic learning and the professional world. It provides students the chance to apply theoretical knowledge in real-world settings and gain insights into industry practices and company culture. For students, internships are invaluable in building connections between what they learn in the classroom and the challenges they'll face in their careers.

The Software Engineering program at the Department of Computer Science and Engineering, Islamic University of Technology (IUT), incorporates a four-month internship as an essential part of its curriculum. This program offers an opportunity for students to combine their academic knowledge with practical experience, ensuring they are well-prepared to meet the demands of the industry. The program's approach allows students to stay updated with emerging technologies while gaining hands-on exposure to industry best practices.

After completing my sixth semester at IUT, I was fortunate to secure an internship at TiCON System Limited, a leading company specializing in Streaming and OTT platforms. This opportunity allowed me to delve into the world of frontend development and gain practical experience in building web applications for a real-world application. As part of my role, I worked closely with a dynamic team and gained hands-on experience in the fast-paced software development environment, particularly focusing on frontend technologies.

## 1.2 Objective

The main goal of this internship was to expand my understanding of frontend development, particularly in the context of real-world applications. Throughout this period, I sought to improve my proficiency in key frontend technologies, such as Next.js, TypeScript, and UI/UX design principles, while contributing to the development of high-quality and responsive web applications. My aim was not just to learn about software development but also to apply this knowledge in creating functional user interfaces that support the business goals of the company.

Another critical objective was to familiarize myself with the industry's best practices by working in a professional environment. By collaborating with senior developers, participating in team meetings, and contributing to the project lifecycle, I gained insight into Agile methodologies, version control using Git, and continuous integration/continuous deployment (CI/CD) practices. These experiences provided me with a comprehensive understanding of how software development works in a team setting.

A significant part of my internship was focused on frontend development. I was tasked with designing and implementing the frontend of a Learning Management System (LMS), which allowed me to apply my academic knowledge to create user-friendly interfaces. Another key responsibility was the frontend development of a mock test application, where I developed the whole frontend aligning with the design from my teammate and integrated the frontend with the backend APIs developed by my teammate. I also performed testing which allowed me to refine my skills in frontend development, API integration, testing, performance optimization, and collaborative development.

The experience gave me the opportunity to develop and strengthen both my technical and interpersonal skills, ensuring that I was not only equipped to build high-quality software but also prepared for effective collaboration in a professional setting.

### 1.3 Scope

During my internship at TiCON System Limited from October 12, 2025, to February 12, 2026, I was primarily involved in frontend development tasks. The scope of my work included the design, development, and testing of user interfaces for web applications. I worked on several projects within the company, contributing to both individual and team-based goals. Some of the key tasks I undertook include:

- **Development of LMS Frontend:** I was responsible for designing and developing the frontend of the Learning Management System (LMS) named Skillscribe using Next.js and TypeScript. My work involved creating reusable components, ensuring responsive design, and optimizing accessibility to cater to a diverse user base. I collaborated with team members where we conducted comparative analysis of established LMS platforms such as Coursera and Udemy to create a user-friendly, intuitive interface that adhered to modern design principles and best practices in UI/UX.
- **Mock Test Application Frontend Development:** As part of my internship, I took the lead in developing the frontend of a mock test application. This involved creating interactive UI elements to provide an engaging user experience. I focused on designing dynamic form inputs, navigation for test sections, and feedback systems to enhance usability. Performance and usability considerations were crucial, as I ensured the application was both fast and user-friendly.
- **API Integration:** A significant part of my responsibilities involved integrating the frontend with the backend API developed by my teammate. I worked closely with him to ensure seamless data exchange between the frontend and backend, which was essential for maintaining a smooth and functional user experience. This task helped me hone my skills in API integration, ensuring that the data flow between systems was efficient and that the application remained responsive and interactive.
- **Testing and Debugging:** In the final phase of my internship, I participated in testing an existing company solution, GPDI. This involved conducting UI testing, API testing, Functional testing and general system validation to ensure reliability and performance compliance. I worked closely with the QA team to identify bugs to improve overall system reliability. This exposure to quality assurance practices helped me better understand the importance of testing in the software development lifecycle and its impact on product quality. I also played an active role in the testing and debugging phases of the two developed projects where I worked with my team to identify and resolve any issues.
- **Cross-functional Collaboration:** Throughout my internship, I collaborated with developers from various disciplines, which provided me with a comprehensive understanding of the web development process. This exposure helped me appreciate how different parts of a project fit together, and how frontend, backend, and design elements work in harmony to produce a complete product. I gained valuable insight into system architecture and the importance of collaboration in achieving project goals.
- **Continuous Improvement:** The internship also involved regular reviews and feedback sessions, where I presented my progress and received constructive feedback from supervisor. This iterative process of improvement allowed me to refine my skills, particularly in frontend development, and contributed to my professional growth. The exposure to real-world development workflows and practices greatly enhanced my technical abilities.

## 1.4 Methodology

The internship followed a structured and collaborative methodology, incorporating elements of Agile and iterative development to ensure continuous improvement and successful project delivery. My role as a frontend developer involved working within a flexible yet organized framework, which emphasized teamwork, feedback loops, and iterative progress. Below are the key aspects of the methodology used throughout my internship:

- **Task Planning and Workflow Management:** During my internship, tasks were distributed and planned using a flexible workflow. The team adopted Agile practices, with sprints planned at the beginning of each week, and progress tracked collaboratively. My responsibilities for the frontend development were outlined clearly at the start of each sprint, with priorities being established based on project requirements and timelines. This method ensured timely completion of tasks while allowing flexibility for unforeseen challenges. We used tools like Google Documents for real-time collaboration and task tracking, rather than formal project management tools.

- **Regular Reviews and Feedback:** Every week, we had sprint review sessions where I presented my progress and demonstrated the features I had developed. The feedback sessions were valuable for understanding the areas where improvements could be made. This feedback loop allowed me to refine my work and align it better with the overall goals of the team.

- **Collaboration and Cross-functional Teamwork:** I actively collaborated with my teammates throughout the internship. For both the projects, I did the frontend development while working closely with designer, fellow frontend and backend developer teammate.Our collaboration focused on ensuring seamless API integration and a consistent user experience across all functionalities. Regular communication between the designer, frontend and backend teams helped identify integration challenges early on and facilitated faster resolution.

- **Agile Development Approach:** Agile practices were especially useful in the development of both the LMS frontend and Mock Test Application. The work was broken down into smaller, manageable chunks, with a focus on iterative delivery. Initially, the core features were developed, followed by performance improvements, user interface enhancements, and integration of new components.

- **Learning and Knowledge Expansion:** Between the completion of the two main projects, I engaged in self-learning to enhance my knowledge in advanced frontend frameworks. I utilized company resources, consulted with senior developers, and attended internal discussions to understand how the company's products were aligned with industry trends. This learning phase helped me deepen my understanding of both the projects I worked on and the broader context of TiCON's business operations.

- **Product Testing and Quality Assurance:** In the final phase of the internship, I participated in the testing of an existing company product, GPDI. I collaborated with the QA team to conduct UI testing, API testing, and functional testing. This allowed me to identify critical bugs, report to the developers to resolve them. Additionally, I worked with my team on debugging and fine-tuning the LMS frontend and Mock Test Application, which were in their final stages of development.

- **Hybrid Work Structure and Communication:** The internship followed a hybrid work structure, with three on-site office days and two remote workdays per week. Communication was maintained primarily through Google Chat and weekly video calls, which ensured that team members could collaborate effectively, despite geographical distances. The remote work model provided flexibility while allowing for close team coordination on all projects.

Overall, the methodology integrated Agile practices, iterative development, and continuous collaboration, enabling me to contribute meaningfully to the projects while enhancing my technical skills. The regular feedback sessions and cross-functional teamwork helped me grow professionally and provided invaluable exposure to real-world software engineering practices.

## 1.5 Limitations

While my internship at TiCON System Limited provided invaluable hands-on experience, it was not without its challenges. The limitations I encountered during the internship, although difficult at times, contributed significantly to my personal and professional growth. Some of the key limitations I faced are as follows:

- **Time Constraints:** The duration of the internship was limited to four months, which constrained the time available to fully explore and understand more complex technologies and tools. Although I gained a significant amount of knowledge and experience, there were instances where certain advanced features or tools could only be superficially explored. The short internship period also meant that I could not gain exposure to longer-term projects or be involved in the entire project lifecycle from start to finish.
- **Scope of Access:** Due to confidentiality and company security protocols, my access to certain proprietary information and internal tools was restricted. This limitation was particularly evident in areas involving advanced software and systems used by the company, which would have provided further insights into end-to-end software development practices.
- **Limited Exposure to Broader Processes:** As an intern focused solely on frontend development, my exposure was limited to the development side of projects. I had little involvement in client-facing interactions, decision-making regarding project scopes, and budget discussions, which would have broadened my understanding of the overall project management process.
- **Cross-Browser Compatibility and Responsiveness:** Ensuring that the applications were mobile responsive and compatible across various browsers was another challenge I faced. Certain user interface (UI) issues became apparent only during testing across different devices and browsers, requiring me to iterate over the frontend design multiple times.
- **Limited Involvement in Backend Development:** Although I was responsible for integrating the frontend with the backend API, my role was limited to the frontend side of the application. This meant I had limited hands-on experience with backend development and database design. While this was in line with my role as a frontend developer, it did restrict my exposure to full-stack development practices. I would have liked more involvement in backend discussions to better understand the complete system architecture.

Despite these limitations, I found the internship experience to be highly beneficial. Overcoming these challenges contributed to my growth as a software engineer. The limitations I encountered helped me develop resilience, adaptability, and problem-solving skills—qualities that will be vital throughout my career. The experience has significantly prepared me for future roles in frontend development and has given me a strong foundation in practical software engineering.

# 2 Company Overview

## 2.1 History

TiCON System Limited (TSL) is a global technology company with its RD center in Uttara, Dhaka, Bangladesh, where I completed my internship as a Frontend Developer focusing on Next.js and TypeScript. Founded in 2007 and headquartered in Seoul, South Korea, with additional presence in the USA, TSL has built a legacy of over 17 years in technology, innovation, and industry leadership. The company operates as a privately held software development firm with an estimated team size of 51-200 employees, primarily skilled professionals in AI, cloud, and software engineering, delivering high-impact solutions worldwide.

## 2.2 Mission

The mission of TiCON System Limited is to drive business innovation through advanced technology solutions. They focus on turning visionary ideas into impactful, real-world results by delivering scalable AI, cloud, and automation systems that create sustainable value for clients, industries, and economies worldwide.

## 2.3 Vision

TiCON System Limited's vision is to harness the power of innovative technologies to empower businesses and individuals with seamless, intelligent, and transformative digital experiences. Their aim is to unlock new possibilities and drive lasting success for businesses through cutting-edge digital solutions.

## 2.4 Services and Products

TiCON System Limited offers a wide range of services to meet the diverse needs of its clients. Some of the key services include:

- **AI Solutions:** Cutting-edge artificial intelligence services, including automation, analytics, machine learning, and content recommendations.
- **OTT Platform:** TiCON provides a feature-rich OTT platform that includes streaming, content management, and monetization solutions for live and on-demand content.
- **Business Intelligence:** Advanced analytics, predictive modeling, and dashboards that enable smarter data-driven decisions and insights.
- **TiCON Cloud:** Secure, scalable cloud solutions powered by AWS, offering cloud migration, optimization, and security services for businesses.
- **Enterprise ERP:** Cloud-based ERP solutions for automating business operations and enhancing workflow efficiency.
- **Cloud Consulting:** Expertise in cloud migration, security, and cost optimization to help businesses optimize their cloud infrastructure.
- **Resource Augmentation:** Providing on-demand skilled professionals for AI, cloud, and software development projects.
- **Custom Software Development:** Tailored software development for unique business needs, from web applications to enterprise solutions.

- **Emerging Technologies:** Exploring disruptive technologies like IoT, Blockchain, and AR/VR to create innovative digital solutions.
- **Software Development:** Full-stack software development services for web, mobile, and SaaS applications.
- **Mobile App Development:** Expertise in building engaging mobile applications across platforms, including iOS and Android.
- **SaaS Development:** Developing Software-as-a-Service (SaaS) products to meet specific business needs with scalable architecture.
- **Custom Payment Processing:** Providing tailored solutions for secure, reliable payment systems.
- **DevOps Services:** End-to-end DevOps services, from architecture review to continuous integration and delivery pipelines.
- **AWS Solutions Architect On Demand:** Providing AWS architecture guidance to ensure high performance and cost efficiency in the cloud.
- **Cloud Cost Optimizer:** Solutions aimed at optimizing cloud resource usage and reducing operational costs.
- **DigitalOcean to AWS Migration:** Helping businesses transition from DigitalOcean to AWS for better scalability and security.
- **Firebase to AWS Migration:** Assisting companies in migrating from Firebase to AWS, improving infrastructure scalability and performance.
- **Blockchain & DApps:** Expertise in building decentralized applications (DApps) and blockchain technology solutions.
- **TICon Agentic AI:** Providing specialized AI solutions through TiCON's own proprietary technology for intelligent decision-making systems.

### 2.5 Clients

TiCON System Limited is trusted by over 100 leading enterprises, startups, and government organizations globally. The company serves clients across multiple industries including Fintech, Edtech, Telecom, Foodtech, Fitness  Healthcare, Media, and more. Notable clients include:

- Airtel
- BRAC Bank
- Robi Axiata
- Unilever
- Binge
- Probasi Bank
- Retina
- Good Bridge
- Kepco KDN

- Rokomari
- Synapse
- TV Storm
- U-Mobile

## 2.6 Tools and Technologies

TiCON System Limited leverages a wide range of technologies to build scalable and efficient solutions. Some of the key tools and technologies used during my internship include:

### 2.6.1 Frontend Technologies

- HTML
- CSS
- JavaScript
- ReactJS
- Next.js
- AngularJS
- Vue.js

### 2.6.2 Backend Technologies

- Node.js
- Python
- Java
- Express.js
- Django

### 2.6.3 Cloud and DevOps

- AWS
- GCP
- Azure
- Docker
- Kubernetes
- Terraform
- CI/CD
- Jenkins

#### 2.6.4 AI and Data Analytics

- Machine Learning
- TensorFlow
- PyTorch
- Natural Language Processing
- Predictive Analytics

#### 2.6.5 Mobile Development

- Swift
- Kotlin
- React Native
- Flutter
- Ionic
- Xamarin

#### 2.6.6 Database Technologies

- MySQL
- PostgreSQL
- MongoDB
- Cassandra
- Firebase

#### 2.6.7 Testing and QA

- Selenium
- Cypress
- Postman
- JUnit
- Mocha
- Chai

#### 2.6.8 Project Management Tools

- JIRA
- Trello
- Asana

#### 2.6.9 UI/UX Design Tools

- Figma
- Adobe XD
- Sketch
- Zeplin

# 3 Company Culture

## 3.1 Working Environment

TiCON System Limited offers a positive and collaborative work environment designed to foster growth, creativity, and teamwork. From the very beginning, the employees were welcoming and supportive, encouraging open communication across all levels. The HR team frequently checked in to ensure that I had the support I needed, and colleagues were always willing to share their expertise and provide assistance when required.
The office space promotes open communication, with no separate cabins for higher management. This open layout allows for easy interactions between team members, senior staff, and management, making it easier for everyone to collaborate. The company maintains a flexible work culture, offering options to work onsite, remotely, or in a hybrid model, depending on personal preference and project requirements.
Team members hold daily stand-ups and share updates to ensure that all team members are aligned. Despite the hybrid working model, the company maintains regular communication with its headquarters in Seoul, South Korea, providing constant updates on ongoing projects.

## 3.2 Company Protocol

TiCON System Limited follows a well-organized protocol that ensures smooth operations within the company. Working hours are from 10:00 AM to 6:00 PM, with punctuality being a key aspect of the company culture. If employees arrive later than the designated time, this is automatically reported through the attendance system.
Daily stand-ups are conducted at the start of each day to track the progress of tasks, review goals, and identify any roadblocks. The company encourages open communication, both online and offline, and promotes a professional yet flexible atmosphere, enabling team members to balance productivity with individual work preferences.
TiCON also offers the flexibility to work in various settings (onsite, hybrid, or remote), depending on project needs and team discussions. The company operates with a collaborative spirit, where everyone, from the management to team members, can engage in productive conversations freely.

## 3.3 Collaboration with Colleagues

TiCON System Limited emphasizes teamwork and collaborative efforts. Employees work in close proximity to each other, fostering an atmosphere of mutual respect and easy communication. The workspace encourages direct collaboration, and colleagues are readily available to assist each other when needed.
While working with my team, I found that the open nature of the office promoted effective communication and knowledge sharing. Project discussions, whether with project managers, directors, or fellow developers, were often held in meeting rooms with large monitors for easier team collaboration and testing.

## 3.4 Events and Activities

Throughout my internship, TiCON System Limited organized several team-building activities and events aimed at strengthening the company's culture and promoting employee well-being. These activities fostered engagement and helped build strong interpersonal relationships among the team members.

Key events included:

- Celebrating the company's founding anniversary with a festive lunch, speeches, and team bonding activities, fostering a sense of unity and appreciation among the staff.
- Organizing a festive lunch on the last day of each month to refresh the mind, provide a relaxed atmosphere, and allow employees to bond and share experiences.
- Hosting farewell celebrations, complete with gift-giving, cake cutting, and heartfelt goodbyes, creating lasting memories and ensuring a warm atmosphere even during departures.
- Eid Reunion celebrations, offering a chance for employees to connect outside of work, celebrate cultural traditions, and share meaningful experiences with colleagues.
- Conducting knowledge-sharing sessions on relevant topics such as AWS, streaming platforms, and OTT systems, which not only promoted professional growth but also encouraged cross-team learning.
- Organizing team-building activities, including collaborative projects such as the school reunion website for the CEO, where team members worked together to meet project goals, enhancing teamwork and coordination.

These events created an inclusive and positive environment, promoting work-life balance and fostering camaraderie among the team members.




**Figure 1:** TiCON on 18th Anniversary






**Figure 2:** Summer Fruit Festival

### 3.5 Intern Policy

Interns at TiCON System Limited were given the opportunity to learn and grow in a supportive environment. During my time as an intern, I received subsidized lunch and snacks, unlimited tea and coffee and the company provided a welcoming gift, including a mousepad and notebook. Interns were encouraged to use their personal laptops but could receive IT support in case of technical difficulties. Interns contributed to important tasks like testing, documentation, and supporting the development process. We were given private Git repositories to commit our code, separate from the company's main repositories, and focused on tasks like product testing and system support.
TiCON System Limited also provided a certificate upon successful completion of the internship, along with additional farewell gifts. The internship policy ensured that interns not only contributed to projects but also had ample opportunity to learn and develop professionally, preparing them for future career opportunities in the software engineering field.

# 4 Company Workflow and Projects

## 4.1 Team-Oriented Development

At TiCON System Limited, the development process follows a dynamic and collaborative Agile approach, fostering seamless teamwork and ensuring continuous integration and deployment (CI/CD). This approach encourages iteration and ensures the project is constantly evolving with ongoing feedback and testing. The development lifecycle typically follows these stages:

- **Client Engagement and Requirement Gathering:** The process begins with initial consultations with the client, followed by signing a Non-Disclosure Agreement (NDA). Once the client's requirements are understood, they are documented as user stories, and the scope of work is established. A contract is signed, formalizing the agreement, and the project proceeds to the planning phase.

- **Planning and Design Phase:** If a design mockup isn't provided by the client, the UI/UX team prepares wireframes and prototypes to visualize the design. These are then added to the project management system, where user stories are prioritized into sprints. If a mockup is already available, the focus shifts directly to refining the design and technical planning.

- **Development Phase:** During development, tasks are assigned based on team members' expertise. The work is typically divided into teams focused on:

  - Frontend Development
  - Backend Development
  - Mobile Development (iOS/Android)

  These teams often work simultaneously, with frontend developers building modular components, while backend developers focus on API development, business logic, and integrating the necessary systems.

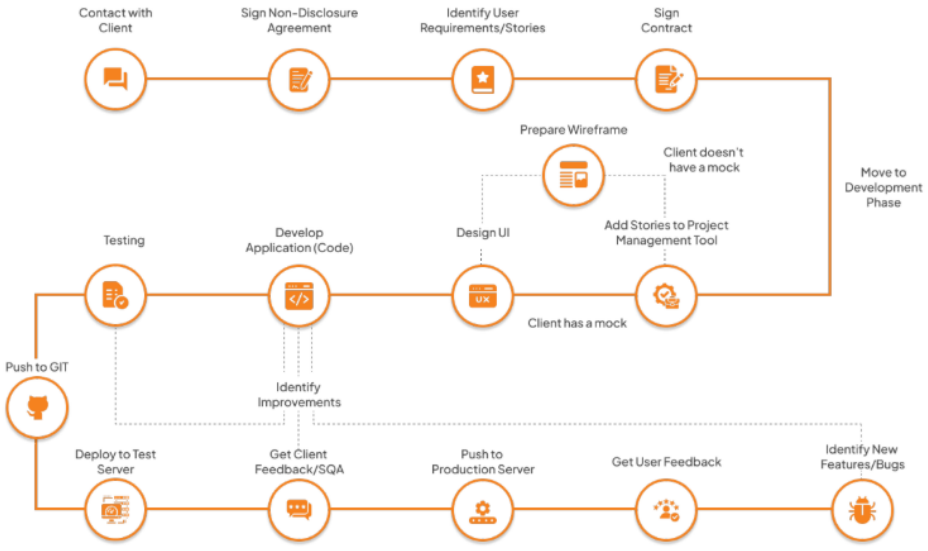



**Figure 3:** Team-Oriented Development Workflow

- **Specialized Engineering Support (AI/ML Integration):** For projects requiring advanced AI capabilities, the AI/ML team collaborates with the development team to integrate AI features such as model training, API connections, and performance optimization. Specialized tasks like computer vision or natural language processing are handled by the AI team.

- **Testing and Quality Assurance:** Dedicated QA teams conduct comprehensive testing including functional testing, regression testing, and validation on staging environments. Once the application is validated, it is pushed to version control, followed by deployment to a staging or test server for further evaluation.

- **Client Review and Deployment:** Following the staging deployment, client feedback is gathered. After making any necessary refinements, the application is deployed to the production environment. Post-deployment, the development cycle continues with iterative updates, including new features or bug fixes based on user feedback.

- **DevOps and Cloud Infrastructure Management:** The DevOps team ensures smooth and automated deployment processes using CI/CD pipelines. They handle infrastructure as code with tools like Terraform and manage the application on cloud platforms such as AWS. Monitoring tools such as Grafana and the ELK stack are used to ensure system performance and observability.

- **Support from Organizational Functions:** The HR department manages employee well-being, onboarding, and attendance monitoring, while the Finance team handles budgeting, project financial planning, and asset management, ensuring smooth operations across the organization.

TiCON's Agile workflow ensures a structured yet flexible development process where collaboration is key, allowing teams to produce high-quality software through continuous feedback and iteration.

### 4.2 In-House Projects

TiCON System Limited has successfully developed and deployed several in-house projects, demonstrating their proficiency in cutting-edge technologies such as AI, IoT, and real-time communication solutions. These projects reflect TiCON's commitment to delivering innovative and tailored solutions to meet the needs of various industries. Some of the key in-house projects include:

- **GPDI Platform:** The GPDI platform (portal.gpdi.ticonsys.com) is an AI-powered procurement intelligence system developed by TiCON System Ltd. It assists companies, particularly Korean enterprises, in discovering, analyzing, and tracking global tender opportunities from major international portals such as sam.gov (US), ungm.org (UN), and ted.europa.eu (EU) in one unified platform. The platform uses AI-powered tender matching to semantically analyze company profiles, products, and services, automatically identifying the most relevant tenders and delivering instant notifications—no manual searching required. Once set up, it continuously monitors opportunities 24/7 and alerts users to high-fit matches, saving time and increasing bid relevance. Key features of GPDI include:

  - AI-powered tender matching
  - Real-time opportunity tracking
  - Automatic notifications and alerts

  The system provides a streamlined process for tender management and offers companies a competitive edge in securing relevant contracts.

- **SkillScribe:** SkillScribe is an AI-assisted digital platform designed to support structured skill development and knowledge documentation. The system focuses on organizing learning resources, tracking user progress, and generating intelligent insights to improve learning outcomes. By integrating automation and data-driven analysis, SkillScribe streamlines the process of skill monitoring, performance evaluation, and content management within a unified environment. The platform enhances learning efficiency, reduces manual tracking efforts, and provides a scalable solution for structured knowledge and skill management. Key features of SkillScribe include:
  - Course management with quizzes, reviews, and wishlist.
  - AI-powered learning assistance and progress tracking.
  - Admin tools for user management, course, and financial reporting.

  SkillScribe serves as an all-in-one solution for individuals and organizations seeking to streamline the skill development process and improve learning outcomes.

- **MaiSafe:** MaiSafe is a comprehensive, enterprise-grade safety management platform developed to address critical safety gaps in industries such as manufacturing and construction. The platform supports compliance with safety regulations like the Serious Accidents Punishment Act (SAPA), providing a 24/7 autonomous safety layer. Key features of MaiSafe include:
  - Automated PPE detection and management
  - Hazard zone monitoring and control
  - Real-time alerts and integrated workflow

  The platform integrates advanced monitoring with a robust backend, ensuring operational safety and reducing risks by automating key safety processes.

- **KDN Project with KEPCO:** TiCON developed an IoT-enabled radiation detection and environmental monitoring system for KEPCO (Korea Electric Power Corporation), one of South Korea's largest electricity providers. The system uses IoT sensors to monitor radiation levels in KEPCO facilities, ensuring the safety of both staff and operations. Key components of this solution include:
  - Radiation detection sensors
  - Data integration
  - Mobile and desktop monitoring

  The project improved KEPCO's ability to monitor environmental safety, reduce human error, and scale the system across multiple facilities.

- **Busan Metro IoT Integration:** Busan Metro, serving over 1.5 million passengers daily, faced challenges in optimizing operations and ensuring smoother, safer travel. TiCON developed an IoT-powered solution that integrated real-time data collection and predictive maintenance capabilities. Key aspects of the project include:
  - IoT sensor integration
  - Predictive maintenance
  - Custom dashboards and analytics

  The solution enhanced safety, efficiency, and reliability, enabling better resource management and providing a more comfortable commuting experience for passengers.

# 5 Team Involvement

## 5.1 Overview of the Team

The team structure at TiCON System Limited includes multiple layers, starting from the senior management, which sets strategic goals and objectives, to the project manager responsible for coordinating team efforts and ensuring deliverables are met. During my internship, the team I worked with was composed of three members from the Islamic University of Technology (IUT): a frontend developer intern (my role), a UI/UX designer intern, a full-stack developer intern and our project manager ( our supervisor). This distribution of roles allowed for specialization in areas like design, backend, and frontend development, yet all tasks were approached collaboratively.
We communicated face to face three offiline work days and on two remote days, we used Google Chat for quick updates, sharing resources, and discussing any issues that arose. Weekly review meetings were scheduled to demonstrate progress, identify any blockers, and receive constructive feedback from the project manager. This agile workflow ensured consistent progress throughout the internship while maintaining clarity and focus in the tasks.

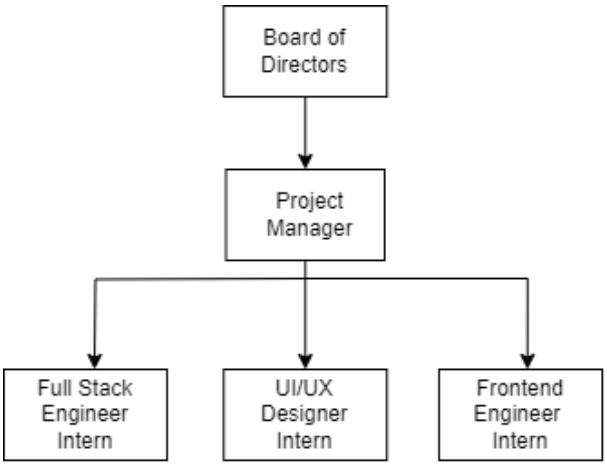



**Figure 4:** Team Structure and Reporting Hierarchy

## 5.2 Responsibilities as an Intern

As a frontend developer intern at TiCON System Limited, I was assigned to work on two major projects: the development of the Skillscribe Learning Management System (LMS) frontend and a mock test application. Below are the responsibilities I held during my internship:

- **LMS Frontend Development:** Initially, I was tasked with the design and development of the LMS frontend. My responsibilities included creating reusable UI components, designing responsive layouts for all pages, and ensuring accessibility features were incorporated. I worked closely with the UI/UX designer intern to transform wireframes into fully functional pages using Next.js and TypeScript.
- **IUT Test Prep Development:** In this project, I developed the frontend for the mock test application, which involved creating multiple interactive components. I worked on ensuring smooth interaction by integrating frontend with backend APIs, a task that required close collaboration with my backend teammate. I also worked on dynamic data binding, performance optimization, and ensuring user-friendly navigation throughout the application.

- **GPDI Testing:** Learned the basics of UI, Functional, API, Integration, Sanity, and Security Testing. Then performed these tests for the GPDI project and reported the bug to the developers to get them resolved to ensure a seamless workflow of the project.
- **Data Entry:** I also assisted in various internal tasks such as data entry for the CEO's school reunion project.
- **Deployment and Error Fixes:** I played an active role in deploying the projects to Vercel, resolving Next.js and TypeScript issues, cleaning up the codebase, and implementing fixes based on supervisor feedback. This included final adjustments to the UI and sorting/filtering functionalities.
- **API Integration and Feature Refinement:** During the second phase of the project, I integrated various backend APIs into the frontend, ensuring proper data flow between the client-side and the backend. My work also involved testing and refining the functionalities based on feedback.

## 5.3 Contributions to Projects

Throughout my internship, I made significant contributions to both the Skillscribe LMS and the Mock Test application projects. Additionally, I contributed to testing the GPDI project and performed essential debugging tasks. Below is a detailed summary of my key contributions:

- **Skillscribe:** I played a central role in designing and developing key features for the Skillscribe LMS. My contributions primarily focused on both the student and admin roles, developing almost all features for the student role and a significant portion of the admin role. For the student role, I developed the navigation bar, footer, profile page, course player, and all course pages, ensuring they were fully responsive. I also integrated advanced functionalities such as the course details page, quiz section, course reviews, wishlist, add to cart page, and a dummy payment process. In addition, I designed and developed the quizzes tab, results page, and certificates section for the course player. I also worked on the learner's dashboard, settings page, community edit functionality, and an AI assistance feature.

  For the admin role, I developed several pages, including User Management, Instructor Verification, Course Management, Category Management, User Report, and Financial Report, as well as FAQ Management, Policy Management, and History pages. I ensured that the website was fully responsive and conducted thorough testing on both the learner-side and admin-side pages, resolving all identified issues.

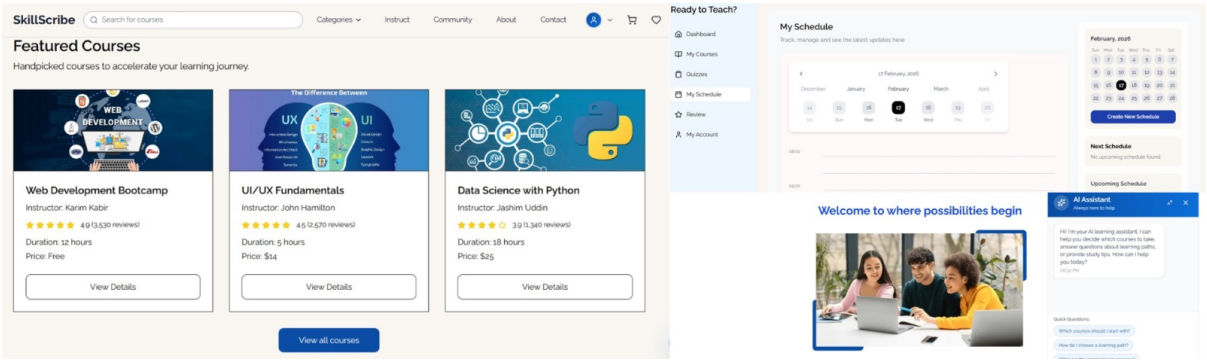



**Figure 5:** SkillScribe (Some Snapshots)

- **IUT Test Prep:** I led the frontend development for the Mock Test application, building pages for both the student and admin roles including the authentication, navbar and footer. For the

student role, I developed pages like the Mock Test, Exam, and Result pages, integrating essential components such as quiz questions, timer functionality, reviews etc. I also worked on the Study Tips and FAQ pages.

For the admin role, I developed the Overview, Contact Messages, Users, and Question Bank pages. I integrated the frontend with backend APIs developed by my colleague, ensuring smooth functionality across platforms. Additionally, I enhanced the UI of the entire project, ensuring it was responsive, aligned with the UI my designer gave me and visually appealing on all devices.

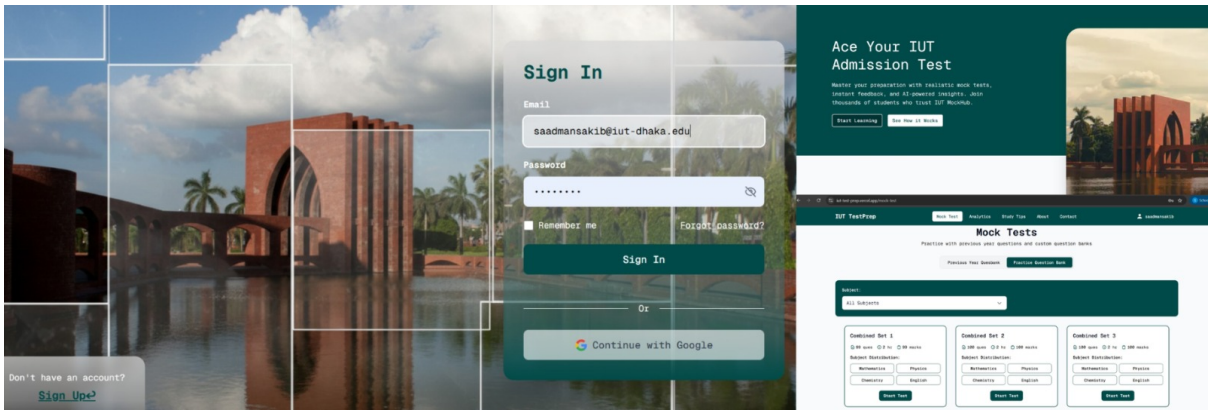



**Figure 6:** IUT TestPrep (Some Snapshots)

- **GPDI Testing:** During my internship, I learned the basics of UI, functional, API, integration, sanity, and security testing. I then performed UI and API testing on the frontend, identifying and resolving various issues, and used tools like Postman for API testing on the GPDI project. My testing experience also extended to functional testing, where I ensured that all features were working as expected. Throughout this process, I gained hands-on experience in testing and debugging, which helped ensure the stability and functionality of the project.
- **UI/UX Improvements and Refinements:** I worked closely with the UI/UX designer intern to ensure the design followed modern user experience principles. I implemented improvements based on feedback from the project manager, ensuring the application was user-friendly, responsive, and visually aligned with the overall design goals.
- **Deployment and Error Fixing:** I helped my teammate in deploying the projects, resolved Next.js and TypeScript errors, cleaned up the codebase, and implemented fixes based on supervisor feedback. I ensured that all pages were responsive and properly aligned across devices. Additionally, I resolved minor build issues and fixed UI-related errors on both the learner and admin sides of the application.

# 6 Learning and Achievements

## 6.1 Technical Skills

### 6.1.1 Languages

During my internship, I honed my skills in the following programming languages:

- **TypeScript:** Used extensively in developing frontend components with Next.js, ensuring type safety and reducing runtime errors.
- **JavaScript:** Applied for frontend tasks, including dynamic content rendering, UI component interaction, and integrating data from backend APIs.

### 6.1.2 Frameworks and Libraries

The following frameworks and libraries were crucial to my development work:

- **Next.js:** Used for building scalable and performant frontend applications, leveraging features such as server-side rendering (SSR) and static site generation (SSG).
- **ReactJS:** Utilized to build reusable and dynamic UI components that were integral to the application's interface.
- **Shadcn/UI:** Implemented for creating reusable, customizable UI components across different parts of the application.
- **Framer Motion:** Integrated to add smooth animations and transitions within the frontend application, enhancing the user experience.
- **React Router:** Used to implement structured and efficient navigation between different pages of the application.
- **Tailwind CSS:** Utilized for quick, utility-first styling that made frontend components both responsive and visually consistent.

### 6.1.3 Databases and Caching

For performance optimization, I worked with:

- **MongoDB:** As the primary database, it stored user data and course information efficiently.
- **Redis:** Used for caching purposes, particularly to store and retrieve frequently used data, which helped in improving performance.

### 6.1.4 AI and API Integration

I integrated various APIs into the frontend to enable dynamic data handling and improve user experience:

- **Apidog and Postman:** Used for mocking APIs and validating API responses during the development phase to ensure smooth integration between the frontend and backend.
- **AI APIs:** Integrated AI features using LLaMA 3.7B models, which were used for data-driven recommendations and functionalities.

#### 6.1.5 Testing

I actively participated in testing the functionalities of the projects:

- **UI Testing:** Performed testing to ensure the user interface was intuitive, functional, and consistent across all pages.
- **Functional Testing:** Ensured that all features on the frontend were functioning correctly, checking for edge cases and user behavior.
- **API Testing:** Conducted API testing using Postman to ensure that the data was properly integrated from the backend and displayed on the frontend.

#### 6.1.6 Version Control and Collaboration Tools

For managing code and collaborating with the team, I used the following tools:

- **Git and GitHub:** Managed version control, ensuring that code was tracked properly and collaborated with teammates seamlessly.
- **Google Chat:** Used for team communication, providing quick updates, sharing resources, and addressing issues that arose during development.

#### 6.1.7 Productivity and AI-Assisted Development Tools

To enhance productivity, I used the following tools:

- **Claude and Grok:** Leveraged these tools to assist in coding tasks and data preparation, improving overall efficiency.
- **ChatGPT:** Used as an interactive assistant for solving development challenges and learning new frameworks and libraries like Next.js and tailwind CSS.

### 6.2 Soft Skills

During my internship at TiCON System Limited, I developed several essential soft skills that were crucial in enhancing both my professional and personal growth. These skills helped me adapt to the work environment, work effectively with my team, and approach challenges with a constructive mindset.

#### 6.2.1 Workplace Discipline and Punctuality

Maintaining workplace discipline and punctuality was essential throughout my internship. I adhered to the office hours, always arrived on time and gave my attendance as well as ensured that I was punctual for meetings, sprint planning sessions, and daily stand-ups. By keeping my project manager informed of my progress and blockers during these sessions, I ensured smooth collaboration. This discipline not only kept the projects on track but also demonstrated my commitment to meeting deadlines and expectations. Moreover, I made sure to manage my time efficiently, balancing my frontend development tasks and testing responsibilities.

### 6.2.2 Critical Thinking

Critical thinking was vital when developing features and resolving issues during the internship. Whether it was debugging frontend behavior, integrating APIs, or ensuring the application was responsive across various devices, I approached every task methodically. I consistently applied logical reasoning to analyze problems and identify optimal solutions. One example was when I faced a challenge in integrating dynamic filters on the course pages. I had to critically evaluate the API responses, ensure they were correctly mapped, and ensure smooth interaction between the frontend and backend. This helped me develop problem-solving skills that were necessary for frontend development.

### 6.2.3 Teamwork and Collaboration

Working as part of a team was one of the most valuable experiences. Collaboration with the UI/UX designer and the backend developer helped me learn the importance of cross-functional teamwork. Whether we were discussing project goals, doing market research, brainstorming solutions for UI/UX improvements, or troubleshooting issues, effective communication was key. I was involved in various discussions and worked collaboratively to resolve design alignments, frontend issues, integrate API data, and improve user experience. The regular sprint planning and progress update meetings kept everyone aligned and provided opportunities for feedback.

### 6.2.4 Attitude and Motivation

Maintaining a positive attitude and staying motivated throughout my internship played a significant role in my overall performance. I approached every task with enthusiasm and was open to learning new tools and technologies. For example, learning how to handle API data and ensuring smooth integration with the frontend was a challenging task, as I never did this before but I stayed motivated to tackle it. This mindset kept me focused and productive, and my enthusiasm for the work was evident in the outcomes I achieved. I was patient and positive throughout the time and embraced each challenge as an opportunity to grow, which helped maintain my engagement and contributed to the team's success.

### 6.2.5 Constructive Feedback

Receiving constructive feedback was crucial to improving my skills as a frontend developer. Throughout my internship, I received feedback from my mentor and project manager during code reviews, which allowed me to refine my coding practices. For instance, after receiving feedback on the responsiveness of the pages I developed, I made necessary adjustments to ensure they were optimized for mobile and tablet devices. I actively sought feedback from my colleagues and mentors to enhance my development approach, ensuring that the final product met both user expectations and technical standards. This continuous feedback loop helped me improve my work and adapt to new challenges.

# 7 Ethical, Social, and Environmental Considerations

## 7.1 Ethical Aspects

Throughout my internship, I placed a strong emphasis on ensuring ethical practices in every aspect of the development process.
For the Skillscribe LMS platform, data privacy and security were of utmost importance, especially when working with sensitive user information like course data, quizzes, and personal profiles.
For the Mock Test Website, I also focused on the secure integration of backend APIs, ensuring that user interactions with the system were safeguarded against common vulnerabilities such as SQL injection, cross-site scripting (XSS), and cross-site request forgery (CSRF). During testing, I was proactive in identifying security vulnerabilities by running various checks using tools like Postman for API security testing and ensuring that all data transfers were encrypted.
During testing of the GPDI platform, an AI-based tender incorporation system, efforts were made to identify potential security vulnerabilities. We intentionally tested for SQL injection patterns and input validation weaknesses to ensure that the system adhered to secure coding standards and professional ethical practices.
Data collection and analytics were designed to be ethical by limiting data retention to the necessary duration and adhering to guidelines for user consent, particularly in light of personal and performance data being captured during quizzes and tests.

## 7.2 Social Aspects

The platforms I worked on had a direct impact on users, especially students, by providing equitable access to educational resources.
For Skillscribe LMS, I designed features that allowed students to engage in structured learning and take advantage of resources that would normally be inaccessible without costly coaching services. The accessibility-focused UI components ensured that all users, regardless of their device or technical proficiency, could use the platform effectively.
One key aspect of the Mock Test Application was to provide a fair and engaging environment for all learners. By integrating features such as dynamic question generation, real-time feedback, and personalized results, I aimed to foster an inclusive environment where all students could track their progress and improve their skills without barriers.
The design and functionality of the Skillscribe LMS also aimed at reducing digital inequality by providing an affordable, scalable, and accessible platform to a broader range of students, including those from less privileged backgrounds. It also aimed to promote fairness by offering all students the same access to the same set of learning resources, removing barriers that might otherwise exist in traditional educational systems.

## 7.3 Environmental Aspects

Although the projects were primarily web-based and thus had minimal direct environmental impact, several design choices were made to minimize the carbon footprint of the system. For instance, I ensured that the frontend code was optimized for performance by using reusable components, minimizing redundant API calls, and reducing the number of unnecessary client-server interactions. This approach not only helped improve the platform's speed and efficiency but also contributed to reducing energy consumption on the user side.
Additionally, during the development of both Skillscribe LMS and the Mock Test Application, I followed sustainable practices by focusing on cloud-based solutions that promoted resource sharing and optimized infrastructure usage. This was particularly important since both projects were hosted on Vercel and Render, which offer scalable, energy-efficient cloud services that automatically adjust resources based on demand. This helped reduce wasted computational resources and, as a result, minimized the environmental impact of maintaining the services.

In terms of sustainability, I worked with my team to modularize the codebase, ensuring that we reused components across various sections of the applications. This modular approach not only contributed to more efficient development and easier future updates but also reduced the need for redundant resources.
While the environmental impact of software development is typically low compared to physical products, adopting these principles ensures that our digital solutions have a minimal carbon footprint.

# 8 Activity Log

**Week 1: 10/12/2025 - 10/16/2025**

- **Date: 10/12/2025**
  We got assigned to an Online LMS Project. We did our market research for the project.
- **Date: 10/13/2025**
  We analyzed the features in detail and made a document.
- **Date: 10/14/2025**
  Designed the learner's wireframes.
- **Date: 10/15/2025**
  Designed the instructor's wireframes.
- **Date: 10/16/2025**
  Designed the admin's wireframes.

**Challenges Faced:**

- Misalignment in design feedback with the team on the first iteration of wireframes.
- Difficulty in analyzing LMS features that catered to all roles (learner, instructor, admin).

**Key Learnings:**

- Gained hands-on experience in using design tools like Figma to create wireframes.
- Learned to communicate effectively with team members to gather design feedback and implement necessary changes.

**Week 2: 10/19/2025 - 10/23/2025**

- **Date: 10/19/2025**
  Completed the remaining wireframes.
- **Date: 10/20/2025**
  UI design of Authentication, Landing Page.
- **Date: 10/21/2025**
  UI design of Landing Page, About Us.
- **Date: 10/22/2025**
  UI design of Contact Us, All Courses.
- **Date: 10/23/2025**
  UI design of All Courses, Course Details.

**Challenges Faced:**

- Difficulty in ensuring all pages were aligned in terms of color schemes, fonts, and spacing.
- Misunderstanding in implementing UI/UX feedback from the supervisor, resulting in multiple revisions.

**Key Learnings:**

- Improved my skills in designing responsive UIs that functioned well across both desktop and mobile devices.
- Gained more knowledge about UI/UX principles and how they directly impact the user experience.

**Week 3: 10/26/2025 - 10/30/2025**

- **Date: 10/26/2025**
  Developed Navbar, Footer components, Hero and CTA sections of the landing page.
- **Date: 10/27/2025**
  Developed Featured Courses, Categories, Testimonials, How It Works, and completed landing page.
- **Date: 10/28/2025**
  Developed All Courses page with dynamic filters and sorting, created routes for courses page.
- **Date: 10/29/2025**
  Implemented Search Functionality, Developed Course Details page, Prepared course data.
- **Date: 10/30/2025**
  Developed Course Player page, Its Resources Tab.

**Challenges Faced:**

- Struggled with designing dynamic filters and ensuring proper sorting functionalities.
- Faced difficulties in ensuring the smooth interaction of UI elements with dynamic data on the courses page.

**Key Learnings:**

- Improved my JavaScript skills by working on dynamic features such as filters and search functionalities.
- Enhanced my understanding of how UI components work together with dynamic data and interactive elements.

**Week 4: 11/02/2025 - 11/06/2025**

- **Date: 11/02/2025**
  Developed Course player Quizzes tab, Quiz page, Prepared quiz data.
- **Date: 11/03/2025**
  Developed Course player Reviews tab, Enhanced the UI of Quiz page, Prepared reviews data.
- **Date: 11/04/2025**
  Developed Course player Certificate tab, Added categories in the courses, made category dropdown, added courses data and new reviews data.
- **Date: 11/05/2025**
  Developed cart components, billing and shipping page, added new quizzes data.
- **Date: 11/06/2025**
  Developed the dummy payment process, wishlist component, wishlist page, added timer in quiz page.

**Challenges Faced:**

- Difficulty in integrating dynamic quiz data and implementing the countdown timer in the quiz page.
- Struggled with developing an intuitive cart component and ensuring a seamless user flow for the payment process.
- Encountered challenges in synchronizing the quiz and review data, leading to display inconsistencies.

**Key Learnings:**

- Learned to implement interactive features such as timers and quizzes that required data binding and performance optimization.
- Gained experience in handling complex UI components like carts and wishlists with dynamic data.
- Improved my problem-solving skills by debugging and refining the integration of dynamic content across pages.

**Week 5: 11/09/2025 - 11/13/2025**

- **Date: 11/09/2025**
  Developed profile page, profile edit functionality, developed the community page and prepared the data.
- **Date: 11/10/2025**
  Developed my learning page, learner's dashboard in profile and the settings page.
- **Date: 11/11/2025**
  Enhanced the UI for certain pages, Developed the AI assistance feature.
- **Date: 11/12/2025**
  Minor enhancements that the supervisor suggested to make.
- **Date: 11/13/2025**
  Made all the pages responsive, enhanced the UI of some pages.

**Challenges Faced:**

- Encountered challenges with aligning different components across various screen sizes to ensure a responsive design.
- Had to work with complex UI features like the AI assistance functionality, which required close attention to detail.

**Key Learnings:**

- Gained experience in building customizable and editable user profiles that improve user interaction.
- Enhanced my skills in creating responsive web designs by ensuring compatibility across multiple devices.
- Improved my ability to implement interactive AI-based features to enhance user experience.

**Week 6: 11/16/2025 - 11/20/2025**

- **Date: 11/16/2025**
  Modularized all pages of the learner into logical reusable components to make it better.
- **Date: 11/17/2025**
  Developed the User Management page for the admin and all the necessary components.
- **Date: 11/18/2025**
  Developed the Instructor Verification page for the admin and all the necessary components.
- **Date: 11/19/2025**
  Developed the Suspend User page for the admin and all the necessary components.
- **Date: 11/20/2025**
  Developed the Policy Management page for the admin and all the necessary components.

**Challenges Faced:**

- Struggled with organizing and modularizing the learner pages to ensure they were reusable across different components.
- Faced challenges in making admin pages intuitive while ensuring they were feature-rich and comprehensive.

**Key Learnings:**

- Gained valuable experience in component-based architecture to ensure reusability across pages.
- Learned how to design and develop robust admin interfaces to help manage users effectively.
- Developed a deeper understanding of data flow between pages and its impact on the user experience.

**Week 7: 11/23/2025 - 11/27/2025**

- **Date: 11/23/2025**
  Developed the FAQ Management page for the admin and all the necessary components.
- **Date: 11/24/2025**
  Developed the Course Management page for the admin and all the necessary components.
- **Date: 11/25/2025**
  Developed the Category Management page for the admin and all the necessary components.
- **Date: 11/26/2025**
  Developed the User Report page for the admin and all the necessary components.
- **Date: 11/27/2025**
  Developed the Financial Report page for the admin and all the necessary components.

**Challenges Faced:**

- Faced challenges in organizing large amounts of data for the admin pages.
- Difficulty in ensuring the UI design remained consistent across all the new pages.

**Key Learnings:**

- Gained experience in developing comprehensive admin pages with complex functionalities.
- Improved my skills in data presentation and integration, ensuring the user interface remained clean and easy to use.
- Strengthened my understanding of how to structure large datasets efficiently on the frontend.

**Week 8: 11/30/2025 - 12/4/2025**

- **Date: 11/30/2025**
  Developed the Chatbot History page for the admin and all the necessary components.
- **Date: 12/1/2025**
  Data Entry Work for CEO's school reunion website.
- **Date: 12/2/2025**
  Data Entry Work for CEO's school reunion website.
- **Date: 12/3/2025**
  More Data Entry Work for CEO's school reunion website.
- **Date: 12/4/2025**
  More Data Entry Work for CEO's school reunion website.

**Challenges Faced:**

- Handling large amounts of data for the CEO's reunion website was time-consuming.
- Ensuring data accuracy and completeness during data entry tasks.

**Key Learnings:**

- Learned how to efficiently manage and enter large sets of data.
- Gained experience in working on multiple tasks simultaneously while ensuring deadlines were met.

**Week 9: 12/7/2025 - 12/11/2025**

- **Date: 12/7/2025**
  Deployed the website on Vercel, solving all the errors of Next.js and TypeScript.
- **Date: 12/8/2025**
  Removed unnecessary codes from the codebase to make it clean.
- **Date: 12/9/2025**
  Made the fixes suggested by our supervisor and changed the UI accordingly.
- **Date: 12/10/2025**
  Added things suggested by our supervisor and fixed minor errors.
- **Date: 12/11/2025**
  Fixed the UI for admin-side pages.

**Challenges Faced:**

- Dealing with deployment-related errors in Next.js and TypeScript, which required careful debugging.
- Ensuring the clean-up process did not result in the removal of any essential functionality.
- Addressing UI inconsistencies and ensuring the designs met the supervisor's expectations.

**Key Learnings:**

- Learned how to streamline codebases by removing unnecessary code while preserving core functionality.
- Improved my skills in frontend UI refinements by closely collaborating with the supervisor to incorporate feedback.

**Week 10: 12/14/2025 - 12/18/2025**

- **Date: 12/14/2025**
  Added "No course found" card in pages where necessary.
- **Date: 12/15/2025**
  Enhanced the UI of All Course player tabs.
- **Date: 12/16/2025**
  Holiday.
- **Date: 12/17/2025**
  Changed the sort and filter options as per the supervisor's suggestion.
- **Date: 12/18/2025**
  Removed unnecessary codes and data.

**Challenges Faced:**

- Ensuring the "No course found" card displayed correctly across all courses.
- Working on improving the UI while maintaining consistency across different sections.
- The challenge of optimizing sorting and filtering functionalities to ensure seamless user interaction.

**Key Learnings:**

- Improved my UI development skills, focusing on user experience and clear messaging.
- Gained experience in refining and enhancing sorting and filtering functionalities for better usability.

**Week 11: 12/21/2025 - 12/25/2025**

- **Date: 12/21/2025**
  Made all the pages responsive.
- **Date: 12/22/2025**
  Tested the learner-side pages.

- **Date: 12/23/2025**
  Tested the remaining learner-side pages.
- **Date: 12/24/2025**
  Tested the admin-side pages.
- **Date: 12/25/2025**
  Holiday.

**Challenges Faced:**

- Ensuring the responsiveness of pages on all screen sizes and devices.
- Testing both learner-side and admin-side pages required careful attention to detail to avoid any overlooked issues.

**Key Learnings:**

- Gained valuable experience in ensuring that applications were responsive and functional across multiple devices.
- Improved my testing skills and attention to detail, identifying issues that needed resolution before final deployment.

**Week 12: 12/28/2025 - 1/1/2026**

- **Date: 12/28/2025**
  Fixed issues found during learner testing.
- **Date: 12/29/2025**
  Fixed additional issues found during learner testing.
- **Date: 12/30/2025**
  Fixed issues found during admin testing.
- **Date: 12/31/2025**
  Holiday.
- **Date: 1/1/2026**
  Brainstormed and researched ideas for the new project.

**Challenges Faced:**

- Identifying and resolving testing issues on both the learner and admin sides.
- Balancing ongoing development tasks while fixing bugs and errors during testing.

**Key Learnings:**

- Improved troubleshooting skills while fixing issues discovered during both learner and admin testing.
- Learned the importance of collaborating with team members during testing to quickly resolve issues.

**Week 13: 1/4/2026 - 1/8/2026**

- **Date: 1/4/2026**
  Analyzed the features of the new project in detail and created a requirement analysis document. Started working on the new project.
- **Date: 1/5/2026**
  Developed the Landing page and the Sign Up page with necessary components.
- **Date: 1/6/2026**
  Developed the Sign In page and About page with necessary components.
- **Date: 1/7/2026**
  Developed the Contact page and half of the Mock Test page. Integrated the backend Auth API and Contact API with the frontend.
- **Date: 1/8/2026**
  Developed the Mock Test page, Exam page, and Result page with necessary components.

**Challenges Faced:**

- Managing multiple tasks simultaneously, such as designing pages and integrating APIs.
- Time constraints due to the need to develop and test multiple pages within the week.

**Key Learnings:**

- Enhanced skills in API integration to ensure a seamless flow of data between the backend and frontend.
- Developed efficient time management strategies to balance coding and integration tasks.

**Week 14: 1/11/2026 - 1/15/2026**

- **Date: 1/11/2026**
  Learned the basics of UI Testing and Functional Testing.
- **Date: 1/12/2026**
  Learned the basics of API Testing and Integration Testing.
- **Date: 1/13/2026**
  Learned the basics of Sanity Testing and Security Testing.
- **Date: 1/14/2026**
  Started UI Testing of the client project GPDI.
- **Date: 1/15/2026**
  Continued UI Testing of the client project GPDI.

**Challenges Faced:**

- Gaining a thorough understanding of different types of testing (UI, Functional, API, etc.).
- Learning how to write effective test cases and manage testing environments.

**Key Learnings:**

- Gained a foundational understanding of various testing methods, including UI, API, and functional testing.

- Learned how to test different components and identify areas of improvement.

**Week 15: 1/18/2026 - 1/22/2026**

- **Date: 1/18/2026**
  Started Functional Testing of the client project GPDI.
- **Date: 1/19/2026**
  Continued Functional Testing of the client project GPDI.
- **Date: 1/20/2026**
  Started API Testing of the client project GPDI.
- **Date: 1/21/2026**
  Continued API Testing of the client project GPDI.
- **Date: 1/22/2026**
  Prepared all the question sets individually and combined them into a CSV for backend integration.

**Challenges Faced:**

- Managing multiple testing phases (functional, API) while ensuring effective feedback to the development team.
- Preparing question sets in CSV format for backend integration, which required careful formatting and organization.

**Key Learnings:**

- Improved testing skills, especially with functional and API testing.
- Gained hands-on experience with API testing tools like Postman and learned to troubleshoot API-related issues.

**Week 16: 1/25/2026 - 1/29/2026**

- **Date: 1/25/2026**
  Developed the Analytics page with its necessary components.
- **Date: 1/26/2026**
  Developed the Profile page with the edit profile option and fixed errors.
- **Date: 1/27/2026**
  Integrated all the backend Exam and Result APIs with frontend.
- **Date: 1/28/2026**
  Integrated all the backend Score Evaluation and History APIs with frontend.
- **Date: 1/29/2026**
  Integrated all the backend AI-related Analytics page API with frontend.

**Challenges Faced:**

- Managing API integration while ensuring the frontend and backend worked seamlessly together.

- Ensuring that data from the backend APIs was presented clearly and accurately in the frontend.

**Key Learnings:**

- Gained experience in API integration and developed a better understanding of data handling between frontend and backend.
- Improved my skills in presenting backend data on the frontend in a user-friendly format.

**Week 17: 2/1/2026 - 2/5/2026**

- **Date: 2/1/2026**
  Developed the Overview and Contact Messages pages with its components.
- **Date: 2/2/2026**
  Developed the Users page and integrated the overview, contact, and user APIs with the frontend.
- **Date: 2/3/2026**
  Developed two tabs of the Question Bank page with its components and integrated the Question Bank API with the frontend.
- **Date: 2/4/2026**
  Holiday.
- **Date: 2/5/2026**
  Developed the Study Tips page and FAQ page for student role with its components.

**Challenges Faced:**

- Balancing multiple pages' development while ensuring smooth integration with backend APIs.
- Ensuring all pages and components were developed with the correct functionality and styling.

**Key Learnings:**

- Developed experience in integrating multiple API endpoints while ensuring consistent frontend behavior.
- Enhanced my ability to manage and implement multiple UI components in parallel.

**Week 18: 2/8/2026 - 2/12/2026**

- **Date: 2/8/2026**
  Developed the Study Tips and FAQ pages for the admin role with its components and integrated backend API with the frontend.
- **Date: 2/9/2026**
  Deployed the website, tested the frontend, and fixed all the build errors and minor errors.
- **Date: 2/10/2026**
  Tested the whole website and fixed all the errors found.
- **Date: 2/11/2026**
  Holiday.
- **Date: 2/12/2026**
  Holiday.

**Challenges Faced:**

- Testing and debugging the website required attention to detail to fix errors and improve performance.
- Managing tasks during the holiday period, while still aiming to complete all assignments.

**Key Learnings:**

- Learned the importance of testing every component to ensure the project is stable and error-free.

# 9 Conclusion

My internship experience at TiCON System Limited has been a transformative journey, significantly contributing to my professional growth and enhancing my understanding of the software development lifecycle. During my time as a Frontend Developer intern, I had the opportunity to apply the theoretical knowledge gained at the Islamic University of Technology (IUT) to real-world applications. This experience allowed me to expand my technical skill set, develop new problem-solving approaches, and understand the intricacies of creating scalable, user-centric applications.

One of the key aspects of my internship was working on the Skillscribe frontend, where I was responsible for designing and implementing dynamic user interfaces using Next.js and TypeScript. I worked closely with talented teammates, learning the importance of cross-functional collaboration and how the integration of frontend, backend, and design elements comes together to create a seamless user experience. The feedback and mentorship from the project manager, along with the team's guidance, played a crucial role in refining my skills and improving the overall quality of the work.

Additionally, my involvement in developing the Mock Test Application and performing testing for the GPDI project provided me with hands-on experience in multiple areas of development. From building reusable components to performing UI and functional testing, I gained valuable insight into the importance of maintaining code quality, the testing process, and the iterative development cycle. I also developed a better understanding of how to approach complex frontend challenges and improve performance while ensuring responsiveness and user-friendliness across all devices.

In terms of soft skills, this internship helped me develop important qualities such as effective communication, teamwork, time management, patience and adaptability. Working with a diverse team, I learned how to collaborate effectively, manage my time, and approach tasks with a positive, growth-oriented mindset. I also gained a deeper appreciation for the significance of constructive feedback in the development process, learning how to accept and apply feedback to continuously improve my work.

The internship at TiCON System Limited has solidified my passion for frontend development and reinforced my commitment to pursuing a career in this field. It has provided me with a well-rounded understanding of both the technical and interpersonal skills necessary to succeed in the software development industry. I am grateful for the opportunity to contribute to meaningful projects, work alongside talented professionals, and gain hands-on experience in a professional setting.